EE 2031 Homework 2

Deadline: 3pm 18th December 2025.

Preparation and submission: You can discuss about the solutions with your friends, but you must solve the questions on physical papers with your own hand-writing. You **must submit** the homework to me **in person,** either during the lectures or by handing it to me when you find me in my office (M2 232). Submissions sent through online channels, including e-mail, will be ignored and will not be graded!

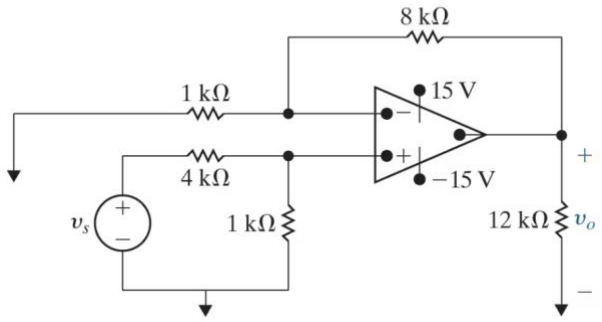
Note: The answers in this HW may not necessarily be Integers. For consistency and precision, please present your results using **three digits after the decimal point** if the result is not Integer.

1) The op amp in the circuit below is ideal.

a) What op amp circuit configuration is this?

b) Find v_o in terms of v_s.

c) Find the range of values for vs so that v_o does not saturate and the op amp remains in its linear region of operation.

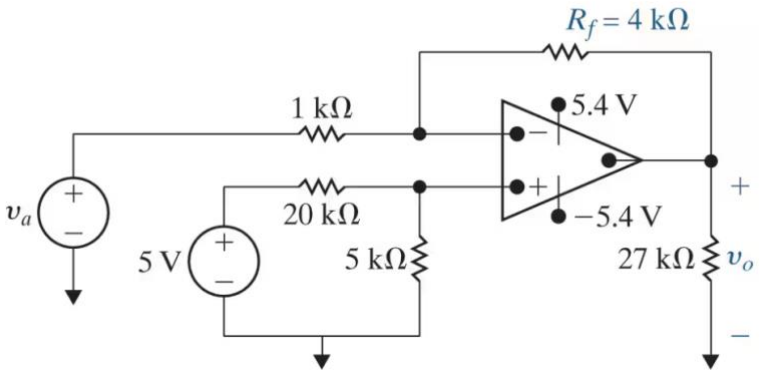


2) The op amp in the circuit below is ideal.

a) What op amp circuit configuration is this?

b) Find an expression for the output voltage v_o in terms of the input voltage v_a.

c) Suppose v_a=2 V. What value of R_f will cause the op amp to saturate?



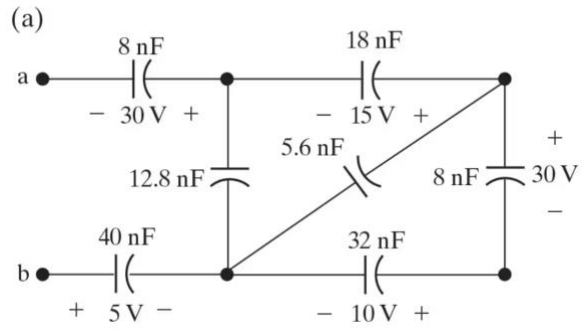
3) The current and voltage for a 0.6 µF capacitor are both zero for t<0. For t≥0, the current is 3cos(50,000t) A.

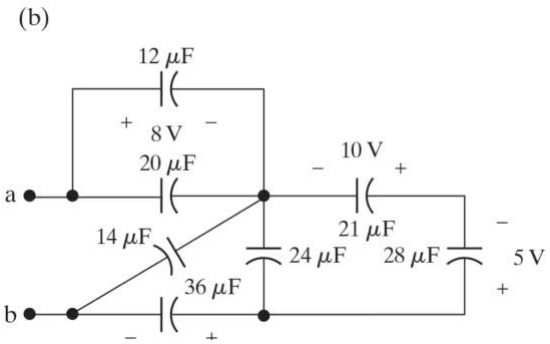
a) Find the expression for the voltage drop across the capacitor in the direction of the current.

b) Find the maximum power delivered to the capacitor any one instant in time.

c) Find the maximum energy stored in the capacitor any one instant in time.

4) Find the equivalent capacitance with respect to the terminals a, b for the circuits shown in (a) and (b).

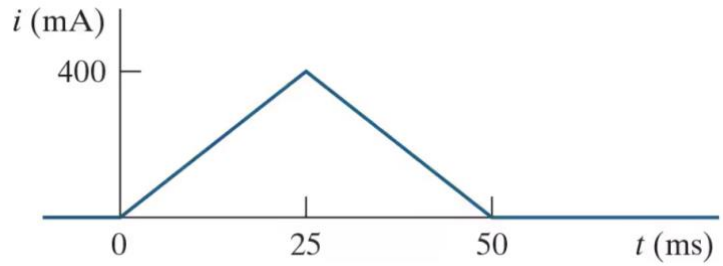




5) The triangular current pulse shown in below is applied to a 375 mH inductor.

a) Write the expressions that describe *i(t)* in the four intervals t<0, 0≤t≤25 ms, 25 ms≤t≤50 ms, and t>50 ms.

b) Derive the expressions for the inductor voltage, power, and energy for each of those four intervals.



6) There is no energy stored in the circuit below at the time the switch is opened.

a) Derive the differential equation that governs the behavior of i_2 if L_1 = 5 H, L_2 = 0.2 H, M= 0.5 H, and Ro = 10 Ω.

b) Show that when $i_g = e^{-10t} - 10$ A, $t \geq 0$, the differential equation derived in (a) is satisfied when $i_2 = 625e^{-10t} - 250e^{-50t}$ mA, $t \geq 0$.

c) Find the expression for the voltage v_1 across the current source.

d) What is the initial value of v_1? Does this make sense in terms of known circuit behavior?

